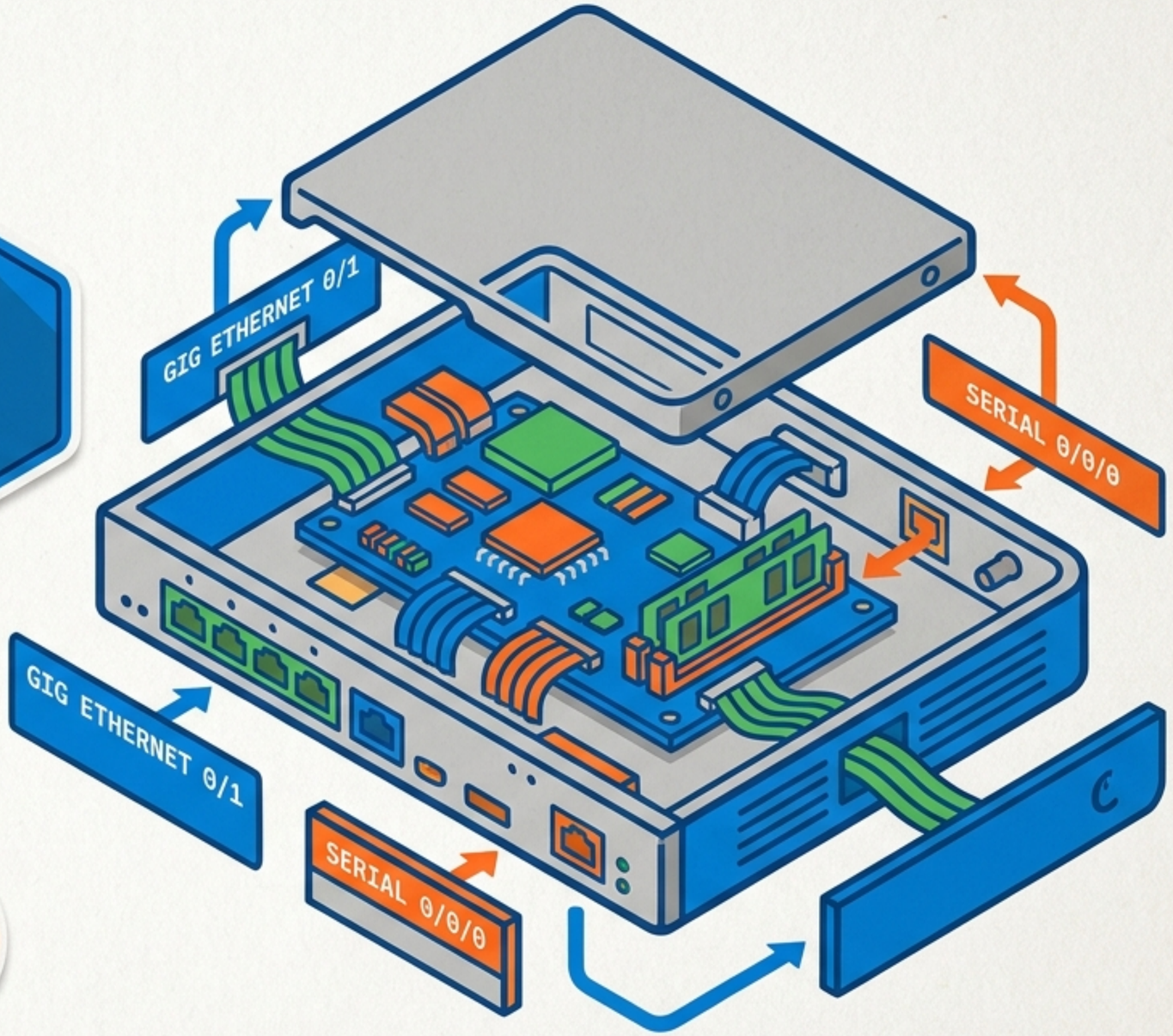
BEGINNER FRIENDLY

100% VISUAL GUIDE

MASTERING RIPv2

A Step-by-Step Field Guide to Connecting Cisco Networks

ZERO CLI INTIMIDATION



NotebookLM

THE PROBLEM: TWO ISLANDS, NO BRIDGE

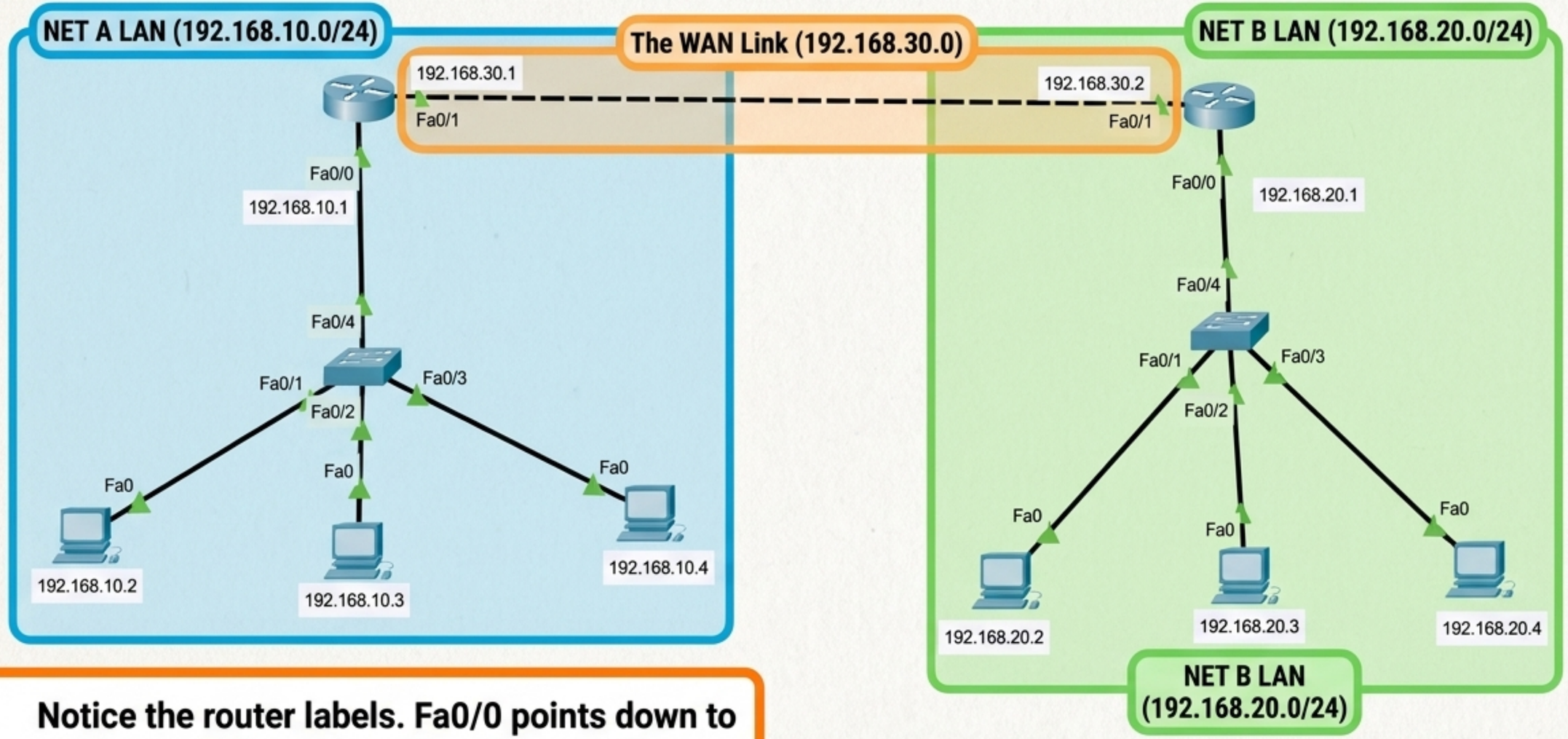


You have two perfectly functioning local networks (LANs). But out of the box, **Router A has no idea Router B exists**. If a PC in NET A tries to talk to NET B, the data is dropped into the void.

The Solution? Routing Information Protocol (RIP). We need to teach these routers how to share their maps.

NotebookLM

Surveying the Territory

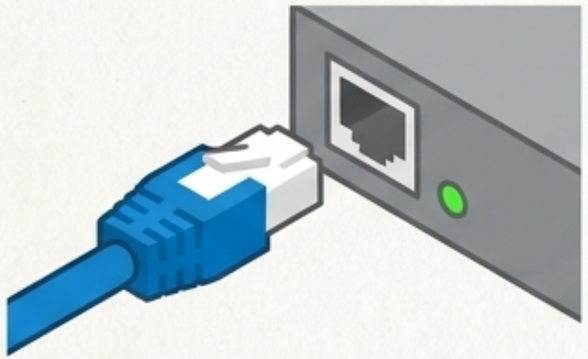


Notice the router labels. Fa0/0 points down to the LANs. Fa0/1 points across to the WAN. These physical cables dictate our code.

NotebookLM

The 3-Step Configuration Recipe

1
The Doors (Interfaces)



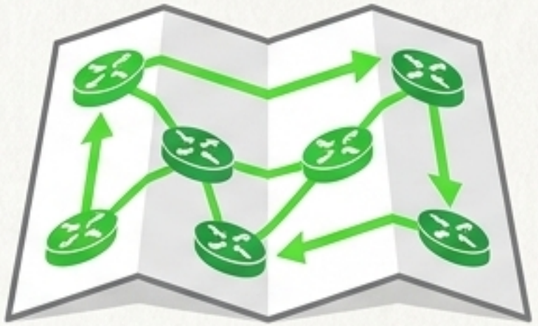
Assign IP addresses to the physical cables plugged into the router (Fa0/0, Fa0/1) and turn them on.

2
The Language (RIPv2)



Turn on Routing Information Protocol Version 2 so the routers can speak the same dialect.

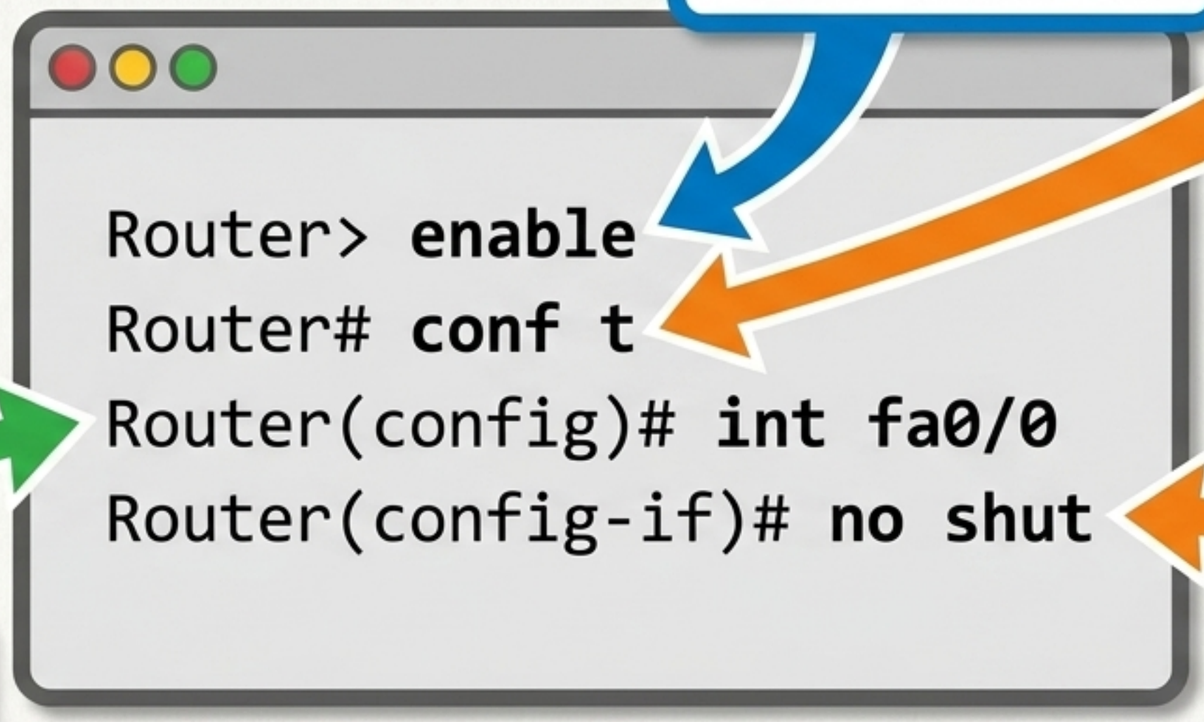
3
The Map (Network Advertising)



Tell each router to broadcast its specific local networks to its neighbor.

NotebookLM

Anatomy of a Router Command



Open the Hood
Enter Privileged EXEC mode.

Grab the Wrench
Enter Global Configuration mode.

Pick a Cable
Target a specific physical port on the router.

Flip the Breaker
Turn the port's power ON—routers default to off!

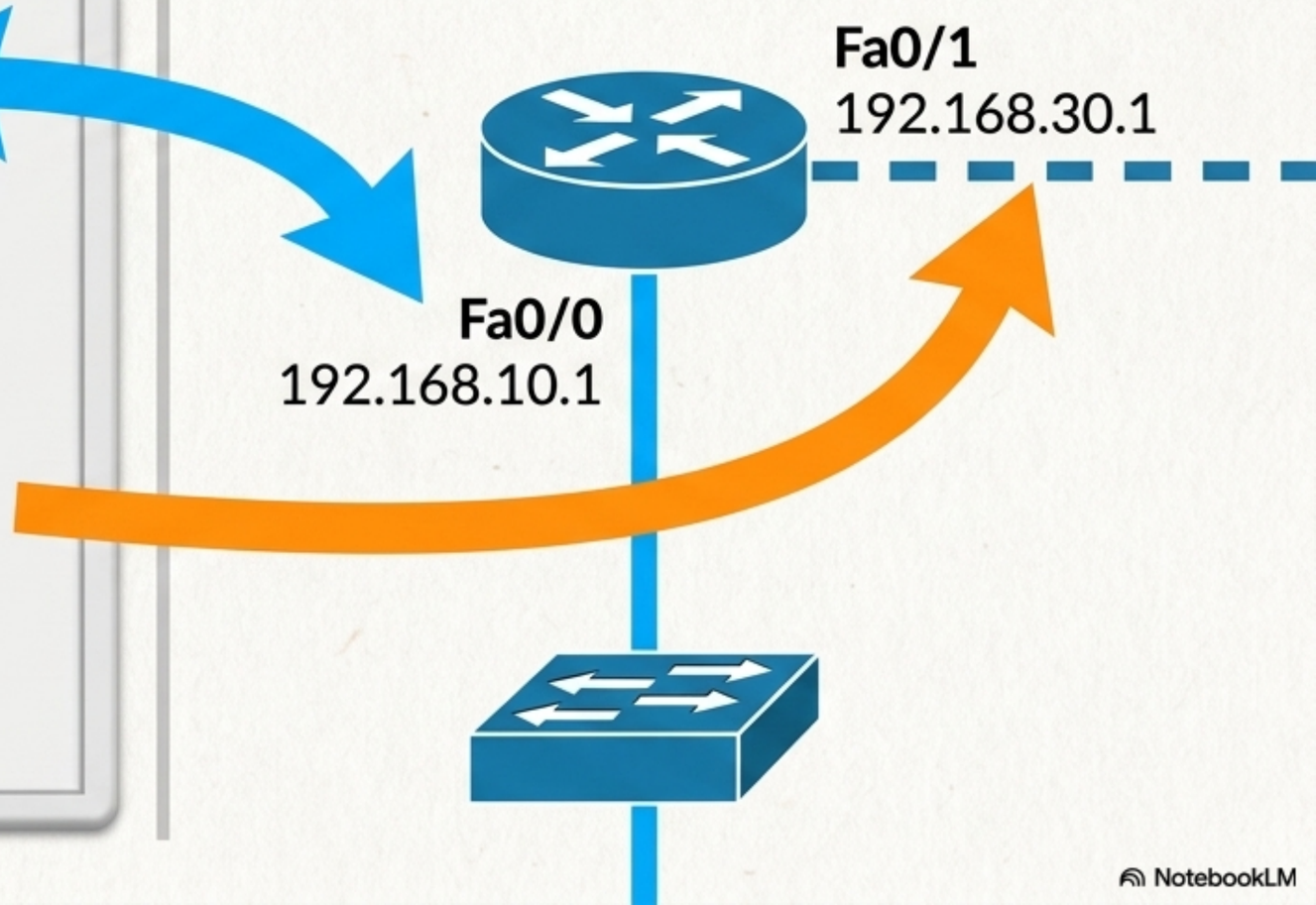
NotebookLM

Wiring Router A (NET A)

The Code

```
Router(config)# int fa0/0
Router(config-if)# ip address
192.168.10.1 255.255.255.0
Router(config-if)# no shut
Router(config-if)# ex
Router(config)# int fa0/1
Router(config-if)# ip address
Router(config-if)# 255.255.255.0
Router(config-if)# no shut
```

The Map



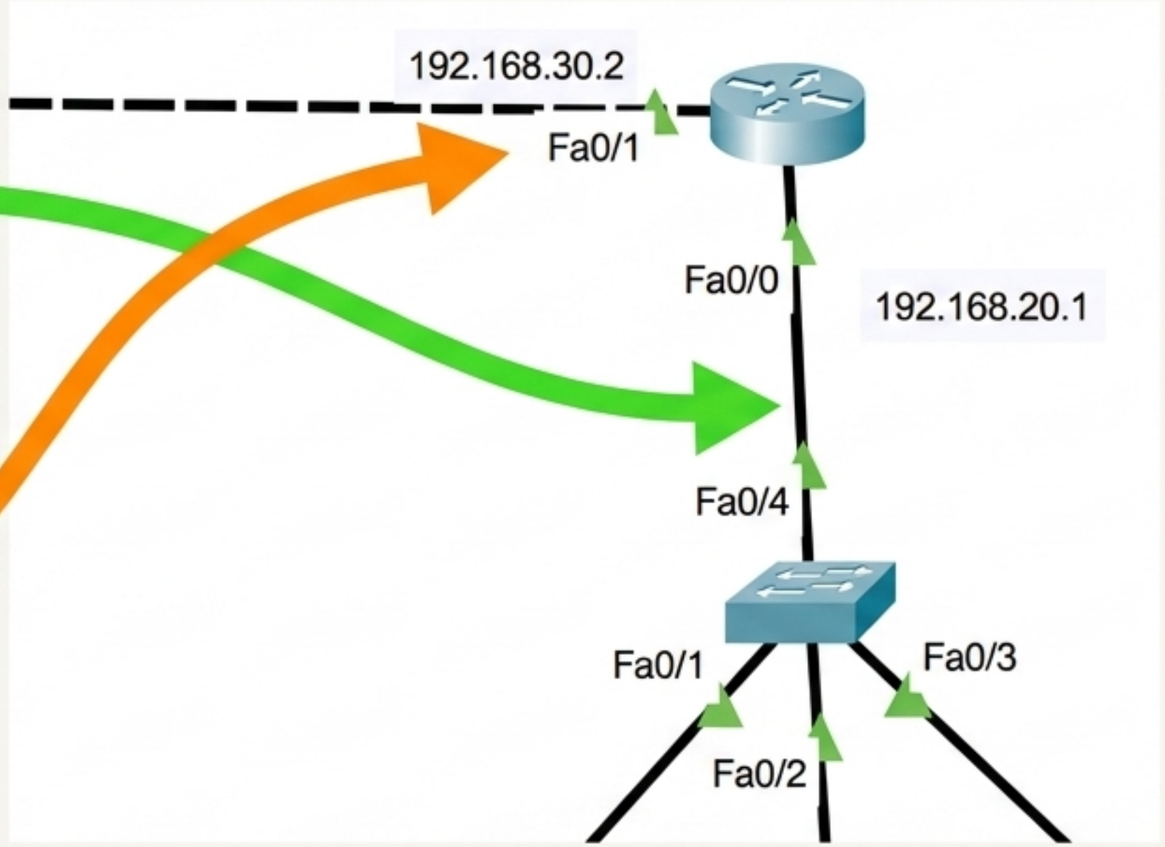
NotebookLM

Wiring Router B (NET B)

The Code

```
Router(config)# int fa0/0
Router(config-if)# ip address
192.168.20.1 255.255.255.0
Router(config-if)# no shut
Router(config-if)# ex
Router(config-if)#
Router(config)# int fa0/1
Router(config-if)# ip address
192.168.30.2 255.255.255.0
Router(config-if)# no shut
```

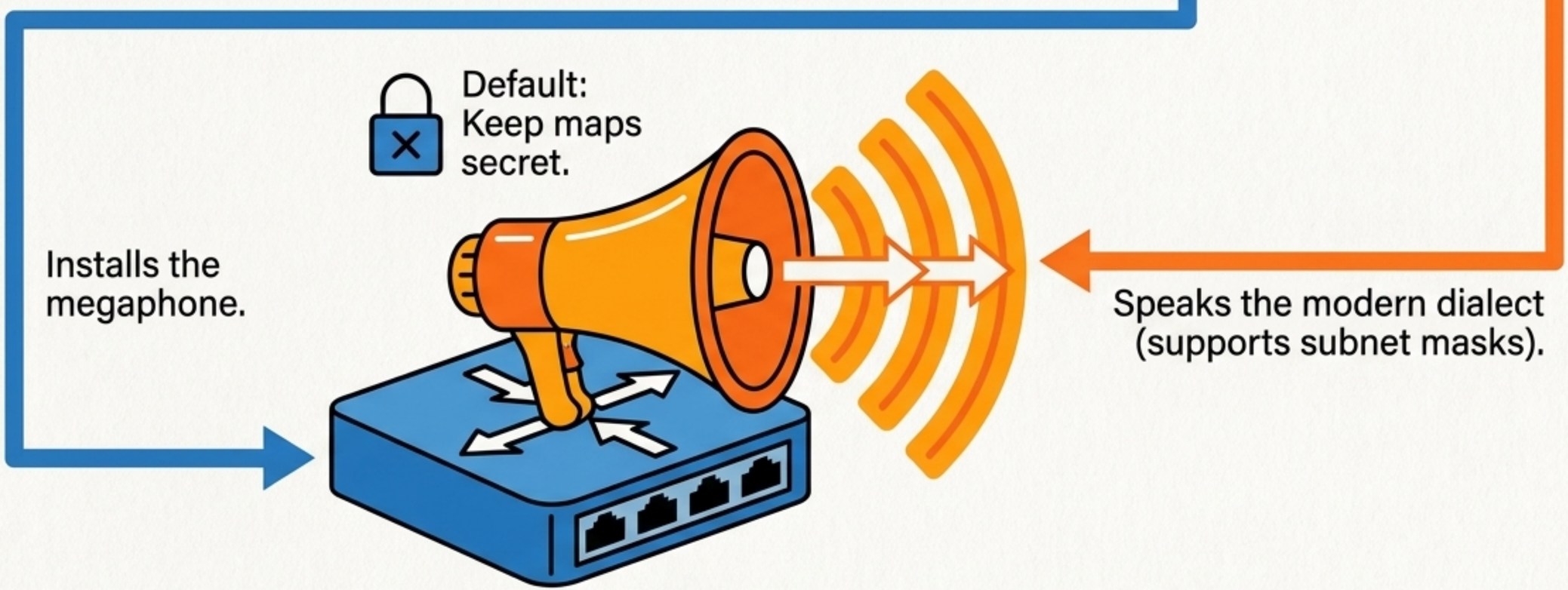
The Map



NotebookLM

Waking Up the Megaphone (RIPv2)

```
Router(config)# router rip
Router(config-router)# version 2
```



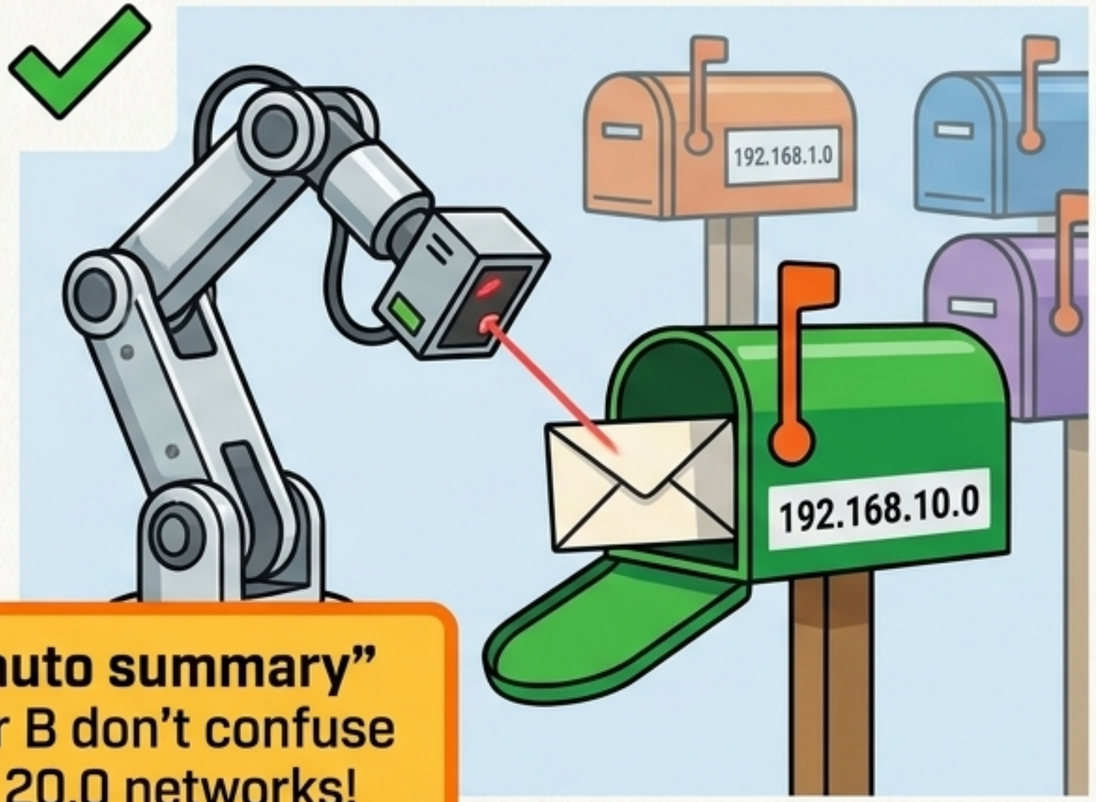
By default, routers keep their maps a secret. Typing "**`router rip`**" installs a megaphone. Typing "**`version 2`**" ensures it speaks the modern, upgraded dialect. Now, they are ready to broadcast.

NotebookLM

The Golden Rule: “No Auto Summary”



Default behavior lazily squashes specific networks into one generic ZIP code. Traffic gets lost.



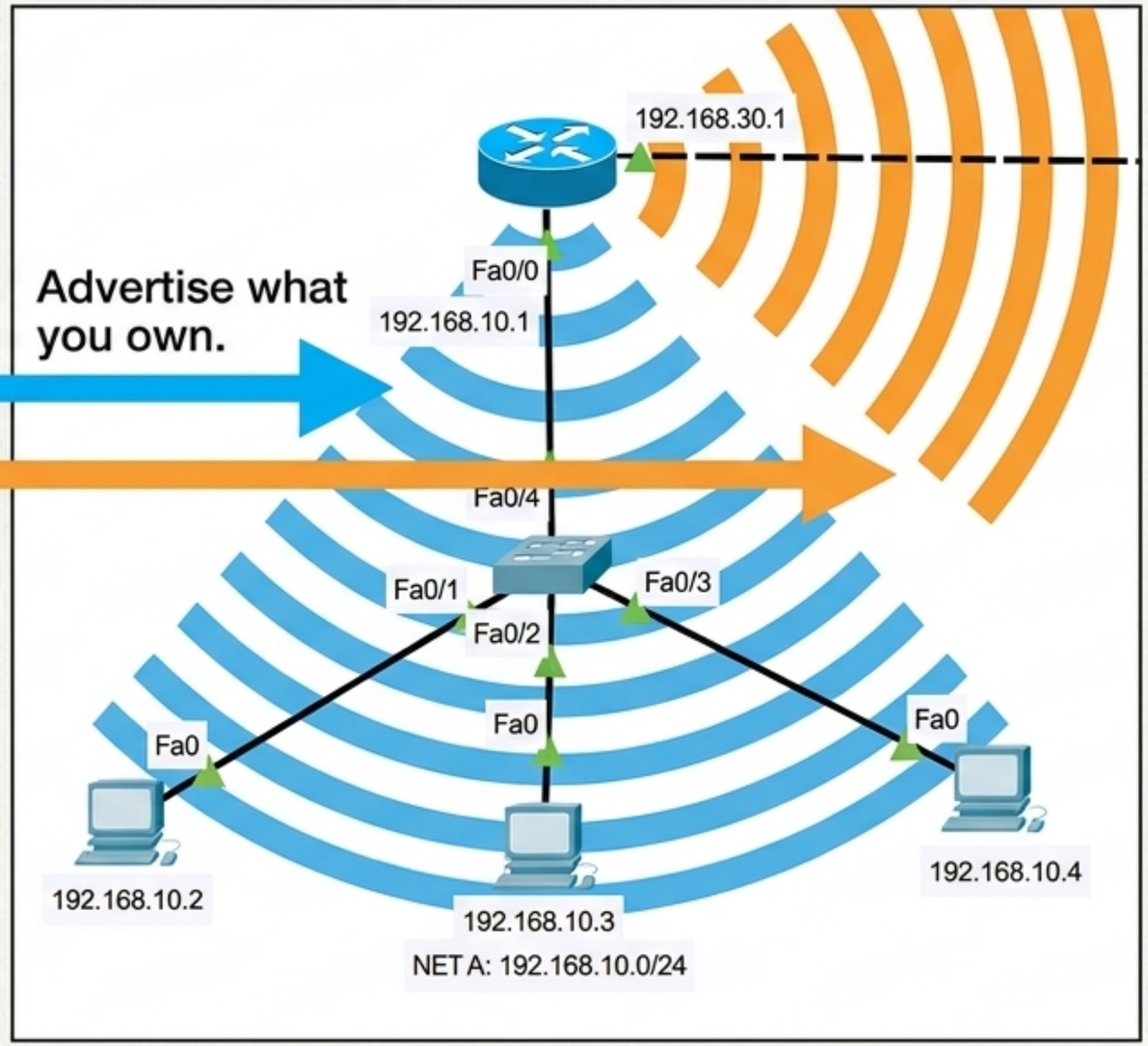
Typing “no auto summary” forces the router to deliver to the exact street address.

Always include “no auto summary” so Router A and Router B don’t confuse their unique 10.0 and 20.0 networks!

NotebookLM

Broadcasting NET A's Map

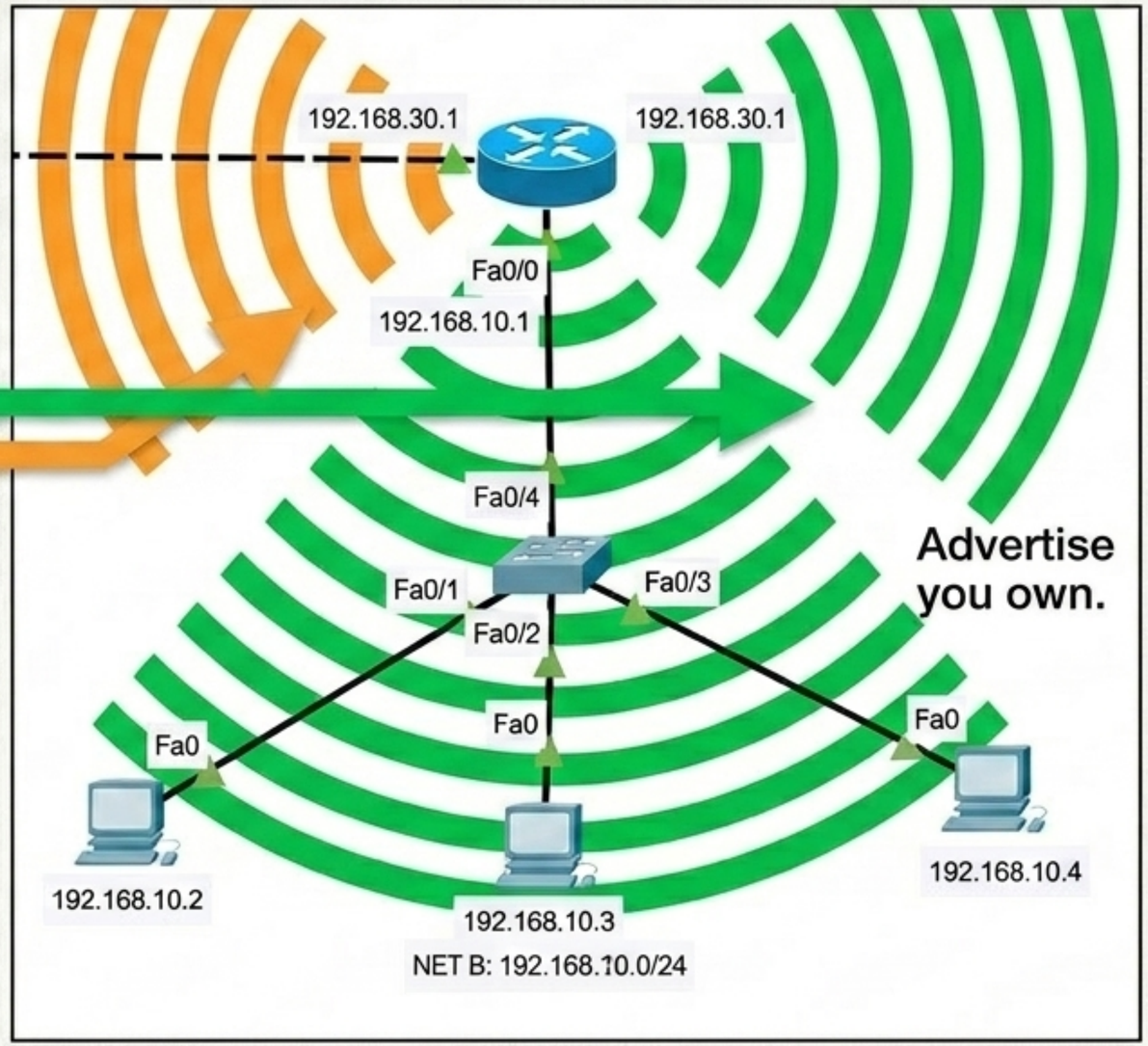
```
Router(config-router)# network 192.168.10.0
Router(config-router)# network 192.168.10.0
Router(config-router)# network 192.168.30.0
Router(config-router)# no auto summary
Router(config-router)# exit
```



NotebookLM

Broadcasting NET B's Map

```
Router(config-router)# network 192.168.20.0
Router(config-router)# network 192.168.30.0
Router(config-router)# no auto summary
Router(config-router)# exit
Router(config)# ex
```



NotebookLM

The Configuration Blueprint

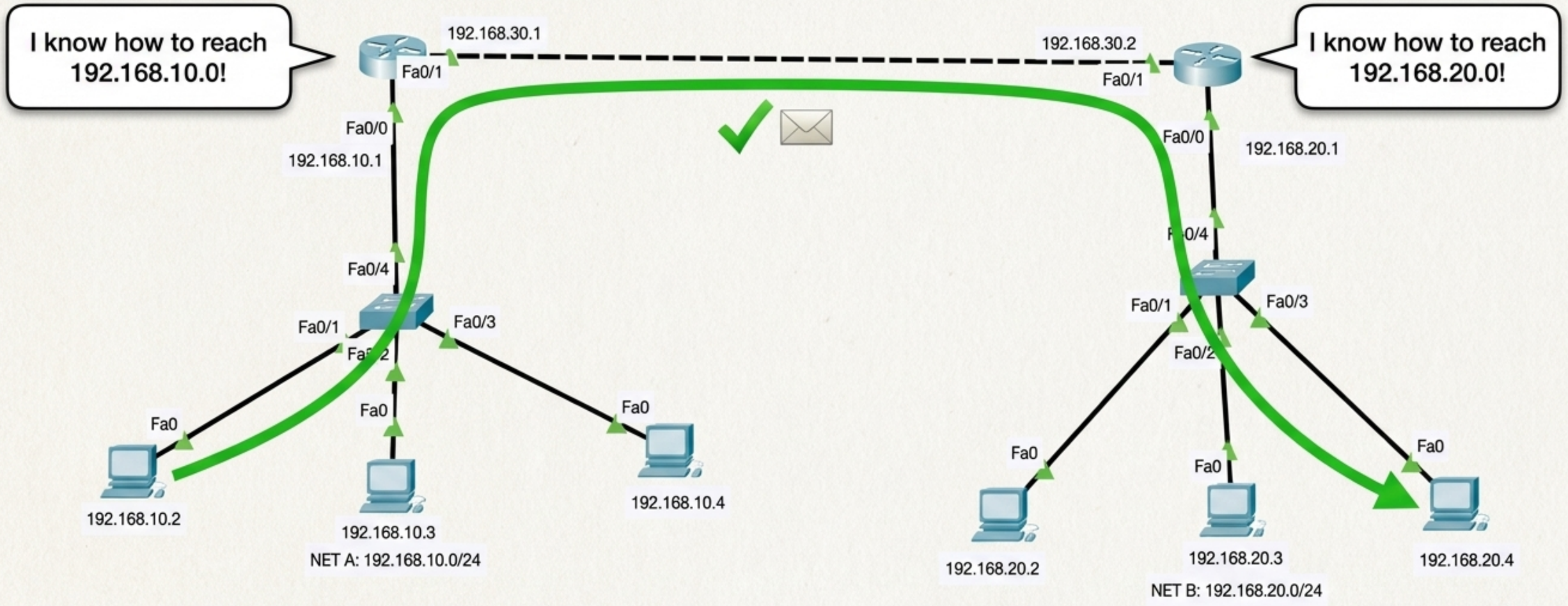
A Tale of Two Routers		
Dimension	**Router A (NET A)**	**Router B (NET B)**
Local Door (Fa0/0)	192.168.**10**.1	192.168.**20**.1
WAN Door (Fa0/1)	192.168.**30**.1	192.168.**30**.2
RIP Advertisements	192.168.10.0, 192.168.30.0	192.168.20.0, 192.168.30.0



Notice the **overlap!** Both routers MUST advertise the shared 30.0 WAN network to shake hands, but they advertise their **own unique** LANs (10.0 and 20.0).

NotebookLM

The Full Picture: RIPv2 in Action



By opening the ports and activating RIPv2, the routers have exchanged maps. The islands are connected. The blueprint is complete.

NotebookLM